QUESTION 3

Predicting ROAS for new player cohorts

A model design for a live-ops game with 2 years of history — what data we need, which methods fit, and which assumptions we must own.

UNIT OF ANALYSIS	PREDICTION TARGET	USE CASE
A new-player cohort (acquired day × channel × country × campaign × platform)	Cumulative ROAS at day-N horizons: D7, D30, D90, D180, D365	UA bidding, channel budget allocation, payback-window guardrails

What data feeds the model

Two years of history gives enough maturity to observe full D365 payback on older cohorts. We build training data at the install level and aggregate for scoring.

1 · UA / cost data

- Spend, impressions, clicks, installs per channel × campaign × ad-set × creative × geo × day
- Blended & network-attributed CPI; bid type (CPI, tCPA, oCPM)
- Attribution source (MMP) and install timestamp

2 · Player-level revenue

- IAP transactions (gross, net, FX-adjusted) per player per day
- Ad revenue at player level (IAA) — eCPM × impressions or platform passback
- Refunds/chargebacks, platform fees, taxes

3 · Player behavior (early signals)

- Sessions, session length, DAU streak on D0–D7
- Tutorial/FTUE completion, level progression, feature adoption
- First-purchase latency, number of purchases by D3/D7

4 · Cohort & context

- Country, device, OS version, language, network
- Game version, live-ops events live at install time
- Seasonality flags, holiday calendar, App Store featuring

5 · Labels (ground truth)

- For mature cohorts (≥ 365 days old): realized cumulative revenue at D7/D30/D90/D180/D365
- ROAS_Dn = cumulative revenue_Dn / acquisition cost
- Right-censor and mark censoring for cohorts younger than the horizon

6 · Data hygiene

- Deduplicate re-attributions; filter fraud (SDK spoofing, click-flood)
- Reconcile MMP installs with store installs; normalise currency daily
- Treat organic and paid separately; document attribution windows

Feature store: one row per install with features computed on snapshot windows (D1, D3, D7).

MODELING APPROACH

Two stages beat one — decouple LTV from cost

Early signals (D0–D7 behavior, cohort, creative) → **Stage A predicted LTV_Dn (regression / survival)** → **÷ CPI by cohort** → **Stage B predicted ROAS_Dn (calibrated, monitored)**

Recommended: Gradient boosting on LTV, then derive ROAS

- Train LightGBM/XGBoost regressors with separate heads for D30, D90, D180, D365 (multi-horizon).
- Use log1p target; Tweedie or zero-inflated loss for heavy-tailed LTV.
- Per-cohort CPI is known at bid time → ROAS = predicted LTV / CPI, aggregated to channel / campaign.

Alt A: Hierarchical Bayesian curve model

- Pool retention + ARPDAU curves by channel × country (partial pooling handles sparse cohorts).
- Projects full LTV curve; gives calibrated uncertainty for budget decisions.

Alt B: Survival + monetisation (two-part model)

- Cox / DeepSurv for churn → expected active days; separate ARPDAU model for revenue per active day.
- $E[LTV] = \int S(t) \cdot ARPDAU(t)\, dt$. Elegant when churn dominates variance.

Start simple (GBM on LTV). Layer uncertainty (Bayesian) and churn structure (survival) where it pays off.

How the model is trained and proven

TARGET

Cumulative revenue per install at horizon N

- Separate heads: D7, D30, D90, D180, D365
- ROAS derived post-hoc: $\hat{y}$(LTV) ÷ realised CPI
- Right-censor young cohorts; use survival-aware loss or pseudo-labels from the D365 model on older cohorts
- Evaluate at the cohort aggregate — that is how UA decisions are made, not at the individual

FEATURES

Snapshotted at D1 / D3 / D7

- Static: country, device tier, OS, language, channel, campaign, creative, bid type
- Early behavior: sessions, min/DAU, levels reached, FTUE finished, retention flags
- Early monetisation: D1/D3/D7 IAP $, ad-impressions/day, first-purchase latency
- Context: seasonality, live-ops event, game version at install, store featuring
- Historical priors: channel × country averages at D30/D90 (leave-one-out)

VALIDATION

Time-based, aggregate, calibrated

- Forward-chaining CV by install month (no future leakage)
- Metrics: cohort-level MAPE / wMAPE on LTV; ROAS calibration curve (predicted vs realised, decile plot)
- Backtest on the last full D365 window; A/B on new bids with a budget guardrail
- Report uncertainty (prediction interval) so UA can size risk — not just a point forecast
- Monitor drift: weekly PSI on features, residual trends by channel × country

Rule of thumb: an early-LTV model that is accurate at D7 cohort level beats a fancier model that over-fits at player level.

ASSUMPTIONS & RISKS

What must stay true — and what could break it

A1 Historical curves generalise

Old cohorts' D90/D365 behavior is informative for new cohorts. Breaks when the game, monetisation, or meta changes materially (new LiveOps, new store, pricing change).

A2 Attribution is stable

MMP attribution windows, SKAN/ATT coverage, and re-attribution rules are consistent. Post-ATT iOS data is noisier: model iOS separately or use conversion-value bucketing.

A3 Early signals correlate with late outcome

D7 behavior is predictive of D365 revenue. For games with long payback, add D30 retraining to capture slower movers.

A4 Cost is known and clean

Network cost at the granularity we score is available, deduplicated, and FX-normalised. Organic attribution errors show up as ROAS inflation.

R1 Risk: Goodhart / feedback loops

Bidding on predicted ROAS re-shapes the data. Keep a hold-out exploration budget (~5–10%) and refresh the model frequently to avoid a collapse of diversity.

R2 Risk: small cells

New geos / creatives have few installs. Use partial pooling or fallback to channel × country priors; never score a cohort without a minimum sample size.

Ship a pragmatic D7-snapshot GBM first · add uncertainty and churn structure where it pays off · re-train monthly, re-calibrate weekly.